XSuM: a generative surrogate of the SBND reconstruction chain trained on the BNB Monte Carlo — technical note

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Abstract

XSuM is a per-event surrogate of the SBND simulation and reconstruction chain: given the true neutrino vertex and the post-FSI final-state particles of an interaction, it samples the record that the cafpyana analysis dataframes hold for that interaction (whether and how many Pandora slices are truth-matched to it, the slice-level block of about fifty reconstructed quantities, and the set of particle-flow objects of each slice), in the dataframe format of the production, so that analyses run unchanged on the surrogated dataframes. The model is trained on the 2026 gen2 BNB + cosmics central-value Monte Carlo (v10_14_02_03) processed with a selection-free cafpyana configuration written for this purpose. This note describes the chain as shipped, its closure against Pandora on held-out files, block by block, and then what is wrong: the two open items of a collaborator's CC1p0pi selection, each traced to its cause. The path to the shipped version is summarised in an appendix.

Contents

1 Purpose and scope **2**

2 Data **2**

3 The shipped chain **5**
3.1 Inputs and the particle encoder 5
3.2 The slice generator 6
3.3 The PFP generator 13
3.4 The surrogated dataframes 16

4 Where we are: the CC1p0pi acceptance **16**

5 What is wrong, and why **17**
5.1 The slice generator: the exit–multiplicity link of CC + pion slices 17
5.2 The PFP generator: the cathode side of the track ends 20
5.3 Smaller items 20

6 Next steps **21**

A The path to the shipped version **21**

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1 Purpose and scope

The physics case is forward folding without re-simulation: a cross-section model that was never run through Geant4, Pandora and cafpyana (another generator, a modified FSI model, a new kinematic region) is transported to reconstruction level by the surrogate and compared with data or with the nominal MC at that level. The surrogate does not fix data/MC disagreement; it reproduces the MC detector response, including its flaws, for new inputs.

Scope: XSuM generates the complete reconstruction record, not analysis quantities. Selection flags, BDT scores and derived kinematics such as $|t|$ are recomputed downstream by the analysers (CC coherent pion: S. Oh; charged kaons: T. Wester; CC1p0pi: M. Fernandez) from the generated variables. Only generator-level truth is an input: the true vertex and the post-FSI primaries. No Geant4 quantity (particle end points, lengths, daughters) enters the models; they are used as a diagnostic reference only (decision 2026-11-07). Generator-internal labels (interaction mode, CC/NC, W, Q^2, E_ν, signal flags) are never inputs; they are kept in a label table for holdout tests. Cosmic-only slices do not depend on the neutrino model and are to be taken from the MC; they are not yet in the surrogated files. Figure 1 is the chain as built; Fig. 2 is where it stands.

2 Data

Production. A cafpyana df-maker configuration (`analysis_village/xsum/`) writes, per flat CAF: `mcnu` (every true neutrino with the full `make_mcdf` truth, the cohpi kinematics and category), `mcprim` (every post-FSI primary), `mcdaught` (the Geant4 daughters of every primary), `slc` (every non-clear-cosmic slice with the cohpi variables and each selection stage as a flag), `pfp` (every PFP of those slices with track, shower and PID variables, kaon PID on all planes, parent links and truth match), `hdr` and POT tables. The training sample is the official SAM definition
`aurora_SBND2026A_gen2_BNBLight_prodgenie_corsika_proton_rockbox0p1_sbnd_CV_v10_14_02_03_flatcaf_sbnd`
(749,262 flat CAFs, 9.94 M triggers, 7.9 TB), processed as 2,000 grid jobs on 2026-10-16 (21.5 s per file, 4,500 CPU-hours): 17.9 M true neutrinos (dirt and out-of-volume included), 5.74 M truth-matched slices. An earlier scratch production (1,633 merged flat CAFs) served the first models and is retired.

A defect of the official production and its repair. The module version used on the grid counted showers among all PFPs of a slice instead of those within 50 cm of the vertex, so the two-track topology never fired and the slice two-track block and the selection flags are empty in the archive. All of them are functions of the PFP table, so they were rebuilt offline exactly as cafpyana computes them (`xsum.prep.repair`) and stored as sidecar `slc` files (the `slcfix` archive next to the production); the training tables and the surrogated dataset use the repaired tables, and so must any comparison with the real chain. The module is fixed for future productions.

Truth–reco link. A slice points to its truth-matched neutrino through `tmatch_idx`; a PFP points to the true particle it was matched to and, through `interaction_id`, to its neutrino. A neutrino can have zero, one or several matched slices; the table keeps the slice with the highest match efficiency and records the count.

Splits. Train / validation / test are 80 / 10 / 10 by input file, never by row. The assignment is a seeded shuffle of the sorted file list and therefore depends on the whole list: tables with different file lists get unrelated assignments. The encoder, the slice flow and the PFP generator are trained on

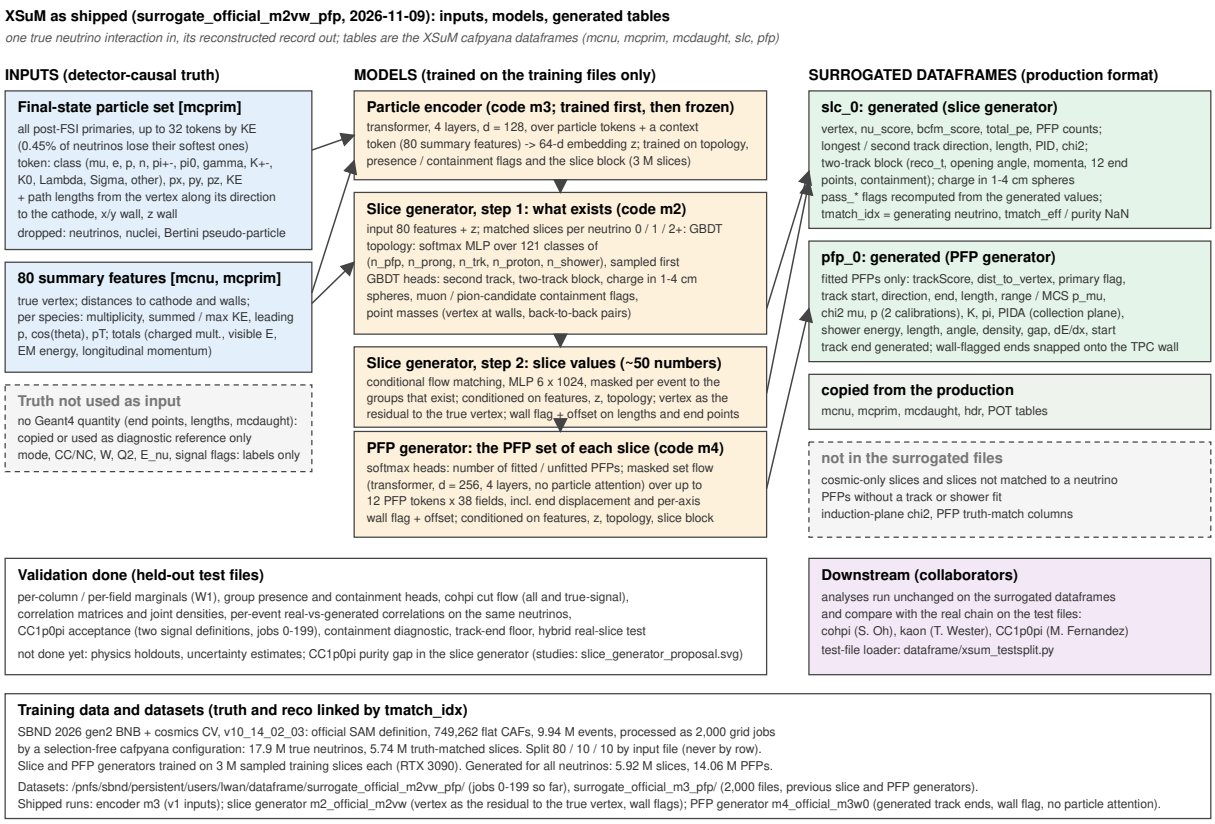


Figure 1: XSuM as shipped: inputs, the chain of models (particle encoder, slice generator, PFP generator), what the surrogated files contain (generated, copied, missing), the validation done and the downstream use.

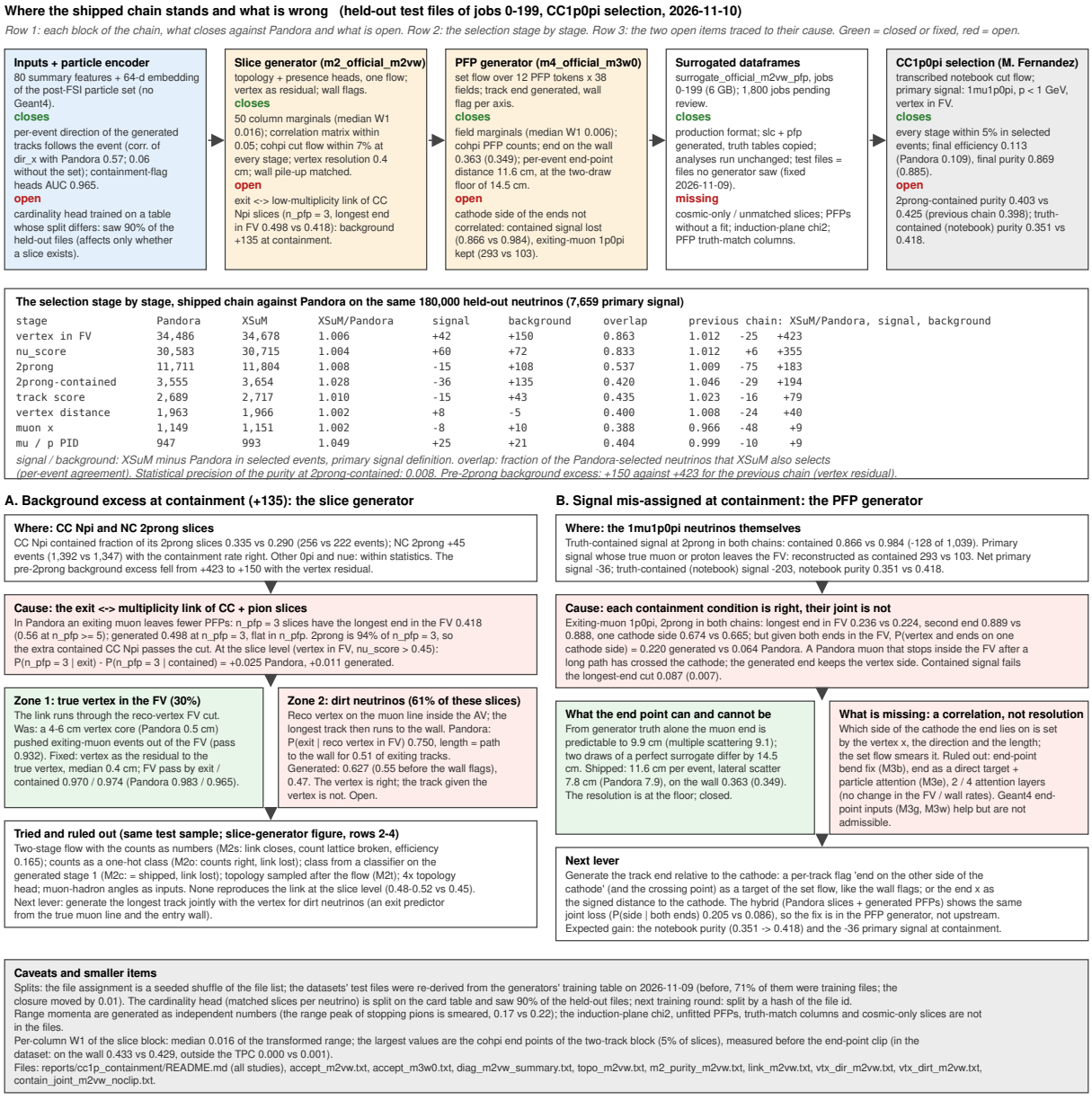


Figure 2: Where the shipped chain stands on the held-out test files: per block what closes and what is open, the CC1p0pi selection stage by stage, and the two open items traced to their cause (Sec. 5).

the matched table (745,363 files) and share one split; the datasets' `file_splits.parquet` had been derived from the generated index (13 files fewer) and the cardinality head uses the 4 M-row card table (740,304 files). Found on 2026-11-09: the datasets' "test" files overlapped the generators' test files at 10–20% only, and 71% of them were generator training files. Since then the split table is derived once from the generators' training table (`xsum.data.splits.file_split_table`, `scripts/make_file_splits.py`; the 240 files with neutrinos but no matched slice are test) and copied into every dataset and the test-file loader (`dataframe/xsum_testsplit.py`). All numbers in this note are on that held-out split unless marked "in-sample" (the variant comparisons of Sec. 5, which share the old selection). The held-out and in-sample closures agree within statistics: the generators do not overfit their training files. The cardinality head still saw 90% of the held-out files (it decides only whether a neutrino gets a slice); the next training round splits every table by a hash of the file id. Holdout tests on physics regions (momentum transfer, energy, angle, multiplicity, TPC) are listed in Sec. 6.

3 The shipped chain

One true neutrino interaction in, its reconstructed record out (Fig. 1). Three models, trained in sequence on the training files of the official sample, one RTX 3090: the particle encoder (code `m3`, run `m3_official`, frozen), the slice generator (code `m2`, run `m2_official_m2vw`) and the PFP generator (code `m4`, run `m4_official_m3w0`). A writer turns the generated tables into production-format files.

3.1 Inputs and the particle encoder

Summary features. 80 fixed features derived from the true vertex and the post-FSI primaries (species, (p_x, p_y, p_z); neutrinos and nuclear remnants dropped): the vertex and its distances to the cathode and the walls; per species (mu, e, p, n, $\pi^{\pm}$, π^0, gamma, $K^{\pm}$, K^0, Lambda, Sigma) the multiplicity above the cohpi thresholds (27 / 30 / 20 MeV for mu / pi / p), the summed and maximum kinetic energy and the leading particle's momentum, $\cos\theta$ and p_T; and totals (charged multiplicity, visible and total kinetic energy, electromagnetic energy, longitudinal momentum). These features carry no azimuth: with them alone the generated track directions were uncorrelated with the event (Pearson +0.06 in x against Pandora on the same neutrinos, Table 2), so the surrogate reproduced the containment *rate* but not *which* events are contained, which is what a truth-defined containment signal tests.

Particle set and encoder. For every neutrino the post-FSI primaries (`mcprim`; neutrinos, nuclear fragments and the Bertini binding-energy pseudo-particle dropped) form a set of up to 32 tokens sorted by kinetic energy (0.45% of neutrinos have more and lose their softest particles). Each token has a species class (12 classes) and (p_x, p_y, p_z, T); it is embedded from the class and from its unit direction, $\log|p|$, $\log T$ and three geometric path lengths from the true vertex along its direction (to the cathode if it heads towards it, to the nearest x/y wall, to the nearest z wall). A context token carries the summary features. A 4-layer transformer ($d = 128$, 4 heads) over the tokens, pooled to a 64-dimensional embedding, is trained on 3 M matched slices for 6 epochs on the targets that matter for selection: the topology class (cross entropy), the group presence and containment flags (binary cross entropy) and the continuous slice block (masked mean squared error); 7 min. The embedding is frozen and appended to the summary features (144 inputs) for both generators. Table 1 shows its effect on the slice heads (the containment flags gain most, muon

candidate AUC 0.892→0.965) and Table 2 on the per-event agreement of the generated directions and end points with Pandora.

head	AUC (v3)	AUC (M3)	log loss (v3)	log loss (M3)
second track	0.995	0.996	0.086	0.082
two-track block	0.999	0.999	0.017	0.016
hits within 1 cm	0.776	0.808	0.330	0.314
hits within 2 cm	0.832	0.860	0.149	0.144
muon-candidate contained	0.892	0.965	0.405	0.235
pion-candidate contained	0.908	0.937	0.288	0.238
back-to-back pair	0.872	0.885	0.087	0.082
topology (121 classes)	accuracy 0.439 / 0.453		2.224	2.138
matched slices per neutrino (0/1/2+)			0.213	0.160

Table 1: Slice-generator heads on the test files with the summary features only (v3) and with the particle-set embedding added (M3; the shipped generators use it).

column	v3	M3
`vtx_x`	+0.91	+0.97
`long_dirx`	+0.06	+0.57
`long_diry`	+0.05	+0.49
`long_dirz`	+0.34	+0.36
`short_dirx`	+0.01	+0.54
`short_diry`	+0.01	+0.46
`short_dirz`	+0.35	+0.38
`muon_end_x`	+0.63	+0.97
`muon_end_y`	+0.58	+0.95
`muon_end_z`	+0.91	+0.96
`cpi_end_x`	+0.95	+0.98
same sign of `long_dirx`	0.53	0.77

Table 2: Pearson correlation, on the same test slices, between the Pandora value and the generated value, without (v3) and with (M3) the particle-set embedding.

3.2 The slice generator

Generation of the slice record proceeds in three steps. (i) A *topology* head: a softmax classifier over the joint pattern of PFP counts (`n_pfp,n_prong,n_trk,n_proton,n_shower`; the 120 most frequent tuples plus "other", whose members are drawn from the training pool), sampled first; its one-hot encoding conditions everything that follows. (ii) Gradient-boosted heads for the presence of the variable groups that may not exist in a slice (second track, two-track block, hits within 1–4 cm of the vertex), for the two containment flags, and for the point masses at detector boundaries (vertex pinned at $z = 0$ or at the x, y walls, back-to-back track pairs). (iii) A conditional flow-matching model (linear path, Gaussian source, 6-layer velocity MLP of width 1024) over the standardised continuous block, masked per event to the groups and dimensions that exist. Positive quantities enter as logarithms, direction cosines and the opening angle through artanh (direction vectors are renormalised), standardisation is robust (median, MAD). Two targets are transformed further. The three reco-vertex coordinates enter the flow as the residual to the true vertex, $\operatorname{asinh}(\Delta/1\,\mathrm{cm})$, and

are added back to the true vertex at generation: as absolute coordinates under a robust scaling the 0.5 cm resolution core was hidden next to the 10 cm-scale misplacements and came out 4–6 cm wide. The longest-track length and the cohpi muon and pion end points carry wall targets, a flag for "at the wall" (length equal to the path from the vertex to the wall along the track direction within 3%; end point on a TPC wall plane) and an offset within the wall band; at generation flagged values are snapped onto the wall, the offset clipped to the band, and an unflagged end point beyond the band is mirrored back inside (`apply_walls`, `tests/test_walls.py`). A separate gradient-boosted head on the full neutrino table gives the number of matched slices per neutrino (0, 1 or 2). Inverse transforms produce a dataframe with the production column names. Trained on 3,000,000 matched slices for 60 epochs (20 min); host memory is kept near 10 GB by splitting the table once on disk into a matched-only table and a 4 M-row subsample for the cardinality head, training the presence heads on 2 M-row subsamples and building the topology one-hot per batch on the GPU.

Closure on the test files. Table 3 gives the presence and containment heads. Table 6 compares every generated column with Pandora (fraction present and Wasserstein-1 in transformed units; median W_1 0.016); Fig. 3 shows the marginals. Table 4 recomputes the cohpi cut flow on the generated and the real test slices: all slices agree within 7% at every stage; the true-signal column (236 neutrinos) ends at 52 against 67, 1.8 standard deviations. Table 5 checks the wall relations of the slice block. Figure 4 compares the Spearman correlation matrix of 19 slice variables on 250,000 test slices: the generated matrix differs from Pandora by 0.008 on average and by at most 0.039 (all 167 defined pairs within 0.05; blank cells are pairs undefined because the two-track variables exist only when `n_trk`= 2). Figure 5 shows joint densities of reco–reco pairs, including the near-deterministic range–length relation, the two-population structure of the vertex charges and of the proton chi2, and the $|t|$–opening-angle anticorrelation that defines the cohpi signal region. Figure 6 shows the conditional the surrogate exists to learn, reconstructed against true quantities on the same neutrinos: the exiting-muon band below the diagonal in range momentum, the mirrored band of muon/pion mis-assignment in the direction, the vertex pile-up at $z = 0$ for out-of-volume interactions, and the two populations of vertex charge.

Group / flag	rate [%]	log loss (marginal)	log loss (GBDT)	AUC
short	60.7	0.6701	0.0819	0.996
two	5.3	0.2059	0.0160	0.999
s1	86.8	0.3896	0.3137	0.808
s2	95.2	0.1922	0.1441	0.860
s3	97.1	0.1308	0.0937	0.883
s4	98.0	0.0979	0.0670	0.899
is_muon_contained	43.1	0.6836	0.2349	0.965
is_cpi_contained	81.3	0.4817	0.2381	0.937
spike_vtx_z	13.0	0.3869	0.0911	0.987
spike_vtx_x	5.6	0.2148	0.0454	0.992
spike_vtx_y	5.4	0.2109	0.0562	0.986
spike_opening_angle	2.5	0.1165	0.0819	0.885

Table 3: Group presence, containment and point-mass heads of the slice generator, test files.

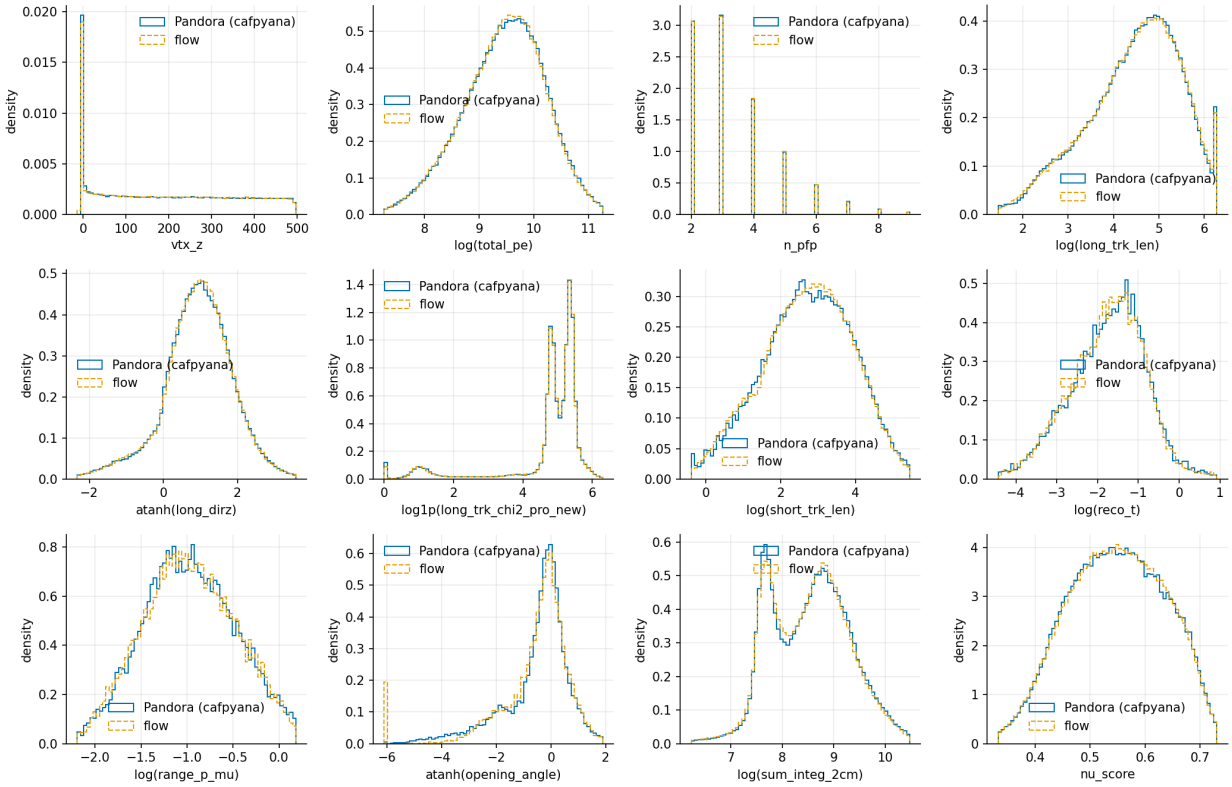


Figure 3: Marginals of generated against Pandora slice variables, test files.

Stage	all, Pandora	all, generated	signal, Pandora	signal, generated
pe_bcfm	236,892	235,915	236	230
2trk	20,116	19,987	123	102
dir	1,118	1,195	100	87
chi2	671	707	91	73
sphere	368	388	84	71
cathode	300	314	72	63
sr	156	152	67	52

Table 4: cohpi cut flow recomputed on real and generated test slices.

all matched slices, test files of jobs 0–199	Pandora	XSuM
longest track: length / path to the wall within ±2%	0.145	0.161
length / path > 1.05	0.076	0.092
cohpi muon end on the wall (±3 cm)	0.429	0.433
cohpi muon end outside the TPC (< −3 cm)	0.001	0.000

Table 5: Wall relations of the slice block in the shipped dataset (`scripts/diag_m2_purity.py`). Without the wall targets these read 0.060, 0.133, 0.079 and 0.187.

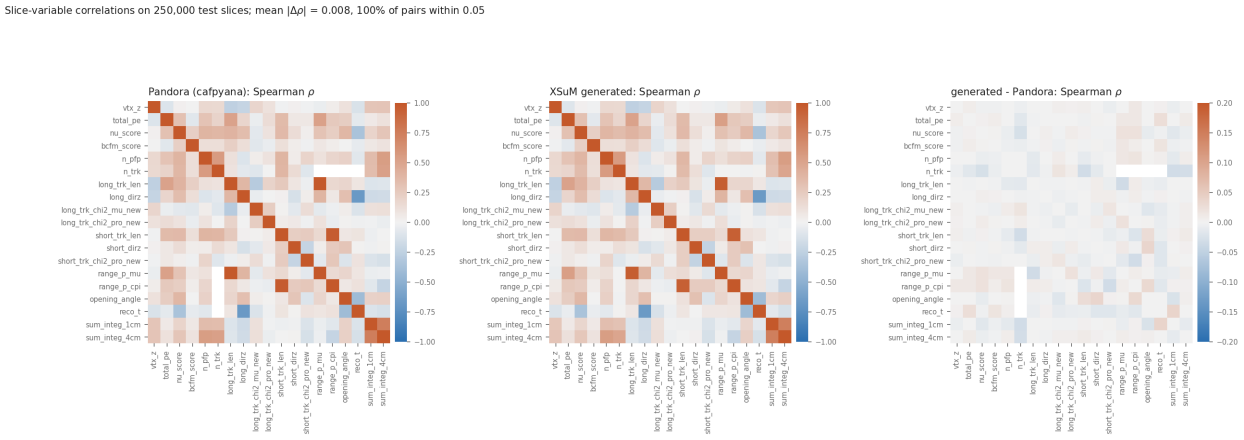


Figure 4: Spearman correlation matrices of slice variables, Pandora (left), XSuM (middle), difference (right, scale ±0.2), test slices.

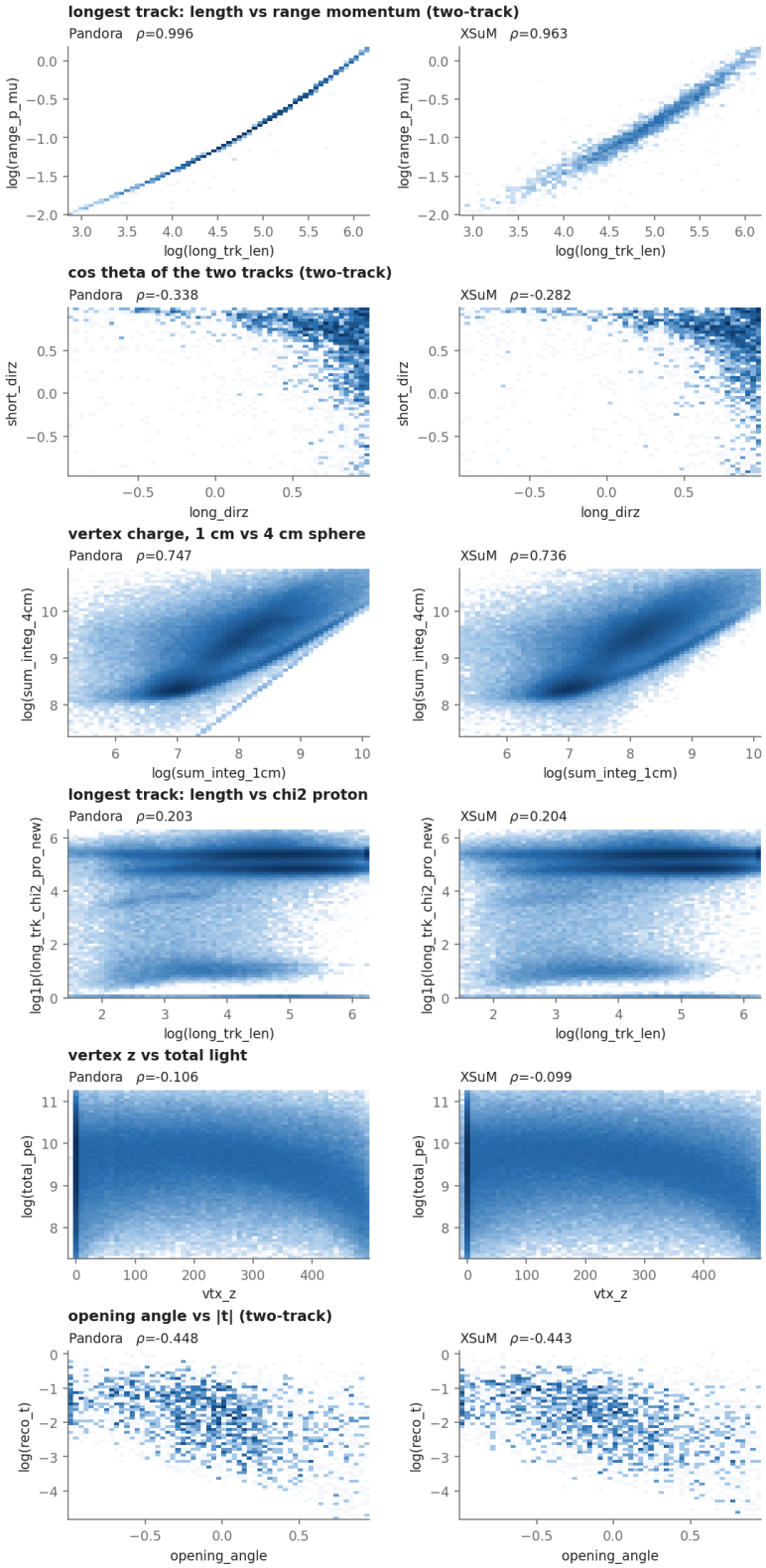


Figure 5: Joint densities of pairs of reconstructed variables, Pandora (left) and XSuM (right), same test neutrinos, same bins and color scale.

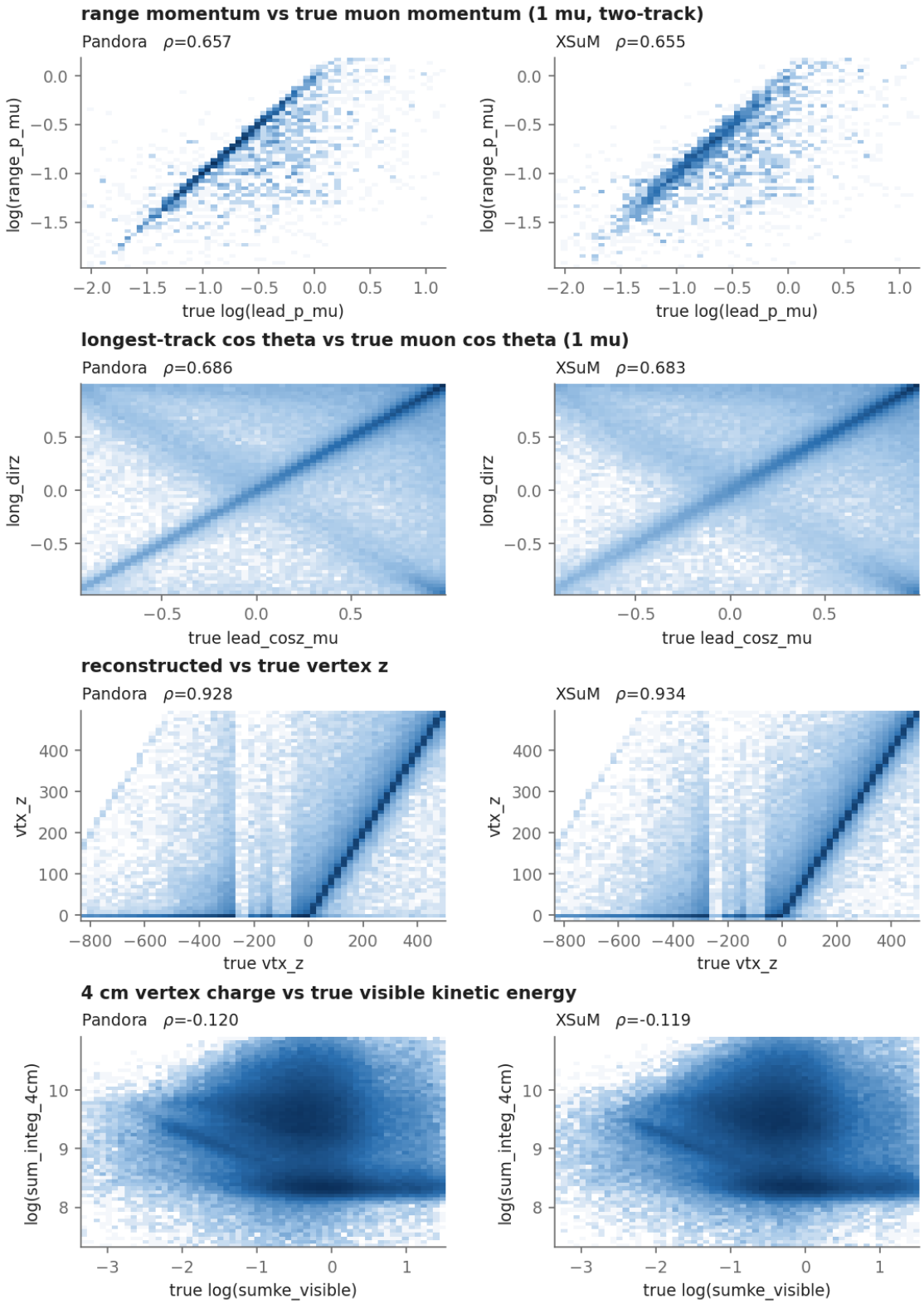


Figure 6: Reconstructed against true quantities, Pandora (left) and XSuM (right), same test neutrinos.

Column	present, real [%]	present, generated [%]	W_1 (transformed units)
n_pfp	100.0	100.0	0.0029
n_prong	100.0	100.0	0.0028
n_trk	100.0	100.0	0.0017
n_proton	100.0	100.0	0.0009
n_shower	100.0	100.0	0.0013
vtx_x	100.0	100.0	0.1338
vtx_y	100.0	100.0	0.3932
vtx_z	100.0	100.0	0.6376
nu_score	100.0	100.0	0.0009
bcfm_score	99.3	100.0	0.0023
total_pe	99.6	100.0	0.0089
long_dirx	99.9	100.0	0.0035
long_diry	99.9	100.0	0.0049
long_dirz	99.9	100.0	0.0074
long_trk_len	99.9	100.0	0.0059
long_trk_pid	99.9	100.0	0.0007
long_trk_chi2_mu	99.9	100.0	0.0079
long_trk_chi2_pro	99.9	100.0	0.0124
long_trk_chi2_mu_new	99.9	100.0	0.0092
long_trk_chi2_pro_new	99.9	100.0	0.0123
short_dirx	60.7	60.8	0.0089
short_diry	60.7	60.8	0.0071
short_dirz	60.7	60.8	0.0138
short_trk_len	60.7	60.8	0.0165
short_trk_pid	60.7	60.8	0.0065
short_trk_chi2_mu	60.7	60.8	0.0183
short_trk_chi2_pro	60.7	60.8	0.0423
short_trk_chi2_mu_new	60.7	60.8	0.0186
short_trk_chi2_pro_new	60.7	60.8	0.0411
reco_t	5.3	5.3	0.0231
opening_angle	5.3	5.3	0.0803
range_p_mu	5.3	5.3	0.0067
range_p_cpi	5.3	5.3	0.0196
muon_start_x	5.3	5.3	3.2344
muon_start_y	5.3	5.3	4.5739
muon_start_z	5.3	5.3	3.6865
muon_end_x	5.3	5.3	1.5167
muon_end_y	5.3	5.3	2.9028
muon_end_z	5.3	5.3	4.0697
cpi_start_x	5.3	5.3	3.1658
cpi_start_y	5.3	5.3	4.5818
cpi_start_z	5.3	5.3	3.7895
cpi_end_x	5.3	5.3	2.5584
cpi_end_y	5.3	5.3	6.2160
cpi_end_z	5.3	5.3	4.8881
sum_integ_1cm	86.8	86.7	0.0112
sum_integ_2cm	95.2	95.3	0.0172
sum_integ_3cm	97.1	97.3	0.0162
sum_integ_4cm	98.0	98.1	0.0193

Table 6: Every generated slice column against Pandora, test files.

The largest W_1 values in Table 6 are the cohpi end points of the two-track block (present in 5% of slices); they were measured at training time, before the end-point clip, and the dataset-level check is Table 5.

Generation. `scripts/generate_slc.py` writes slice records for any truth table: slice cardinality from its head, then topology, presence, point masses and the flow, then the production column names and the recomputed cohpi flags; about 12,000 neutrinos per second on one GPU, 27 min for the 17.9 M neutrinos of the official sample (5.92 M slices).

3.3 The PFP generator

The particle-flow objects of a slice are generated as a set. Per matched slice the fitted PFPs (those with a track or shower fit; the few without either are counted separately) are sorted by distance to the vertex and padded to 12 (99.99% of slices have at most 12). Each PFP carries 38 fields: presence flags (track fit, shower fit, MCS fit, primary), distance to the vertex, track score, start point, direction, range and MCS muon momenta, chi2 for muon, proton (both calibrations), kaon and pion on the collection plane, PIDA and the chi2 degrees of freedom, the shower energy, length, opening angle, density, conversion gap and dE/dx, and the track end point: the displacement end−start (3 numbers, generated independently of the direction, so that multiple scattering can bend the chord away from the initial direction), the excess of the track length over the chord, and per axis a flag that the end lies on a TPC wall plane with the outward offset from it; flagged ends are snapped onto the wall at generation. Flags are dequantised, fields that do not exist for a PFP are filled with noise and restored from the flags after generation. The conditioning vector (309 numbers) is the truth features with the embedding, the topology one-hot and the generated slice block, so the PFP set is consistent with the slice record. Two softmax heads give the number of fitted and of empty PFPs; a masked set flow (transformer velocity net, $d = 256$, 4 layers, no PFP-to-particle attention) generates the records. Trained on 3,000,000 slices for 40 epochs (45 min).

Closure on the test files. The number of fitted PFPs is predicted with 93.2% accuracy (log loss 0.173 against 1.552 marginal): the slice block already carries the multiplicities. Table 7 is the physics closure: the cohpi multiplicities (prongs, track candidates, proton candidates, showers) recomputed from the generated PFPs against the same recomputation on the real PFPs; the exactly-two-tracks topology rate is 8.07% generated against 7.97% real (7.97% from the slice table itself). Table 9 compares every generated field with Pandora (median W_1 0.006), Fig. 7 the marginals.

Track end points. The end point of the longest track against Pandora on the same neutrinos: 11.6 cm per event (3D median; the straight line start + length × direction 10.0 cm), lateral deviation from the straight line 7.8 cm (Pandora 7.9), end in the FV 0.486 (0.497), on a wall within 3 cm 0.363 (0.349), outside the TPC 2.2% (0.2%). Signed residuals are below 0.5 cm on every axis: a resolution, not a bias. How much of it is intrinsic? Using Geant4 truth only as a reference (`scripts/diag_end_floor.py`), for contained muons below 1 GeV the best straight-line prediction from the vertex, direction and range misses the Geant4 end by 9.9 cm (median; 1.6 cm from straggling and interactions in the length, 9.1 cm from multiple scattering in the direction), and two independent draws of a perfect surrogate differ by 14.5 cm (median, 90% quantile 40 cm). The shipped per-event distance is at this floor; the target is the distribution, not the event. PFP-to-particle cross-attention was tested and left out (Table 8): two layers gain 1 cm and reach the floor, four add nothing, and the end-in-FV and on-wall rates do not change in any generator-truth class, while the parameters and the training time double.

Count	mean, Pandora	mean, generated	0/1/2/3/4, Pandora	0/1/2/3/4, generated
n_prong	1.587	1.596	4344 / 161854 / 97884 / 28326 / 6499	4199 / 161344 / 97982 / 28245 / 6652
n_trk	1.439	1.444	18228 / 165064 / 90105 / 22177 / 4166	18529 / 164416 / 89836 / 22416 / 4290
n_proton	0.409	0.408	194597 / 90844 / 13320 / 1559 / 159	195310 / 90076 / 13168 / 1729 / 184
n_shower	0.374	0.383	222945 / 51455 / 19369 / 5085 / 1262	223121 / 49851 / 19657 / 5973 / 1478

Table 7: cohpi multiplicities recomputed from PFP sets, test slices (counts of slices with 0 / 1 / 2 / 3 / 4).

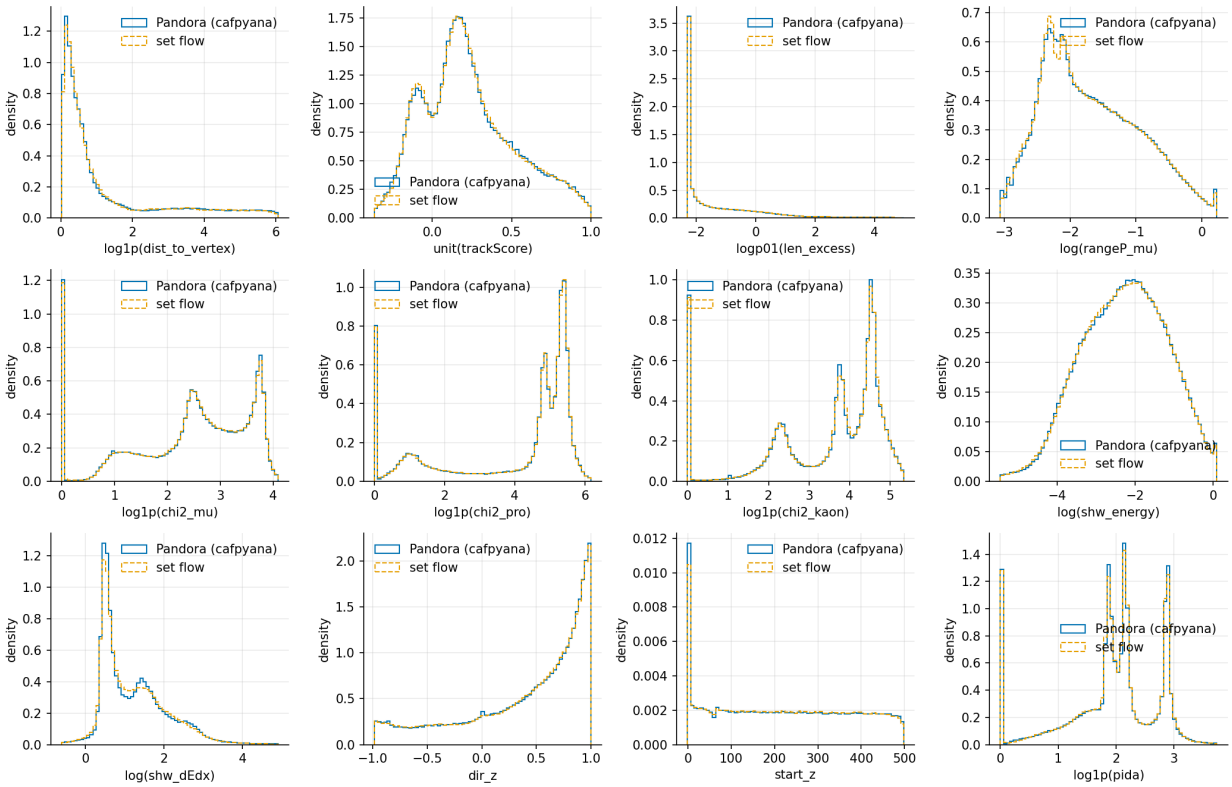


Figure 7: PFP fields, generated set against Pandora, test slices (valid tokens).

Field	Pandora median (flags: mean)	generated	W_1 (transformed units)
has_trk	1.000	1.000	–
has_shw	0.999	0.999	–
has_mcs	0.707	0.704	–
is_primary	0.927	0.931	–
dist_to_vertex	0.549	0.552	0.0175
trackScore	0.191	0.191	0.0019
start_x	-18.933	-18.983	0.5232
start_y	1.692	1.899	0.5254
start_z	221.122	221.744	0.7106
dir_x	0.008	0.010	0.0021
dir_y	0.005	0.005	0.0017
dir_z	0.604	0.605	0.0024
len_excess	-1.908	-1.902	0.0066
rangeP_mu	-1.792	-1.796	0.0023

Field	Pandora median (flags: mean)	generated	W_1 (transformed units)
mcsP_mu	-0.916	-0.932	0.0165
chi2_mu	2.628	2.628	0.0057
chi2_pro	4.894	4.897	0.0035
chi2_mu_new	2.629	2.627	0.0062
chi2_pro_new	4.906	4.905	0.0045
chi2_kaon	3.924	3.930	0.0056
chi2_pion	2.954	2.952	0.0060
pida	2.032	2.034	0.0065
chi2_ndof	3.332	3.299	0.0234
shw_energy	-2.234	-2.240	0.0039
shw_len	3.128	3.130	0.0050
shw_open_angle	0.026	0.027	0.0036
shw_density	1.593	1.597	0.0037
shw_conversion_gap	0.539	0.544	0.0170
shw_dEdx	0.973	0.981	0.0147
disp_x	0.005	0.010	0.0026
disp_y	0.006	0.003	0.0048
disp_z	0.844	0.850	0.0109
wall_x	0.055	0.052	–
wall_y	0.062	0.059	–
wall_z	0.098	0.102	–
wall_off_x	-0.077	-0.075	0.0107
wall_off_y	0.103	0.221	0.1417
wall_off_z	0.225	0.219	0.0144

Table 9: Every generated PFP field against Pandora, test slices.

particle attention layers	0 (shipped)	2	4	Pandora
longest-track end, |gen−real| median [cm]	15.3	14.3	14.2	14.5 (floor)
straight line start + len·dir [cm]	13.0	12.8	12.8	
lateral deviation from the straight line [cm]	9.0	8.8	8.6	7.9
end in the FV	0.470	0.469	0.472	0.497
end on the wall (±3 cm)	0.355	0.363	0.359	0.349
validation loss	0.750	0.734	0.733	
parameters	2.2 M	4.4 M	5.5 M	
training time	7 min	20 min	25 min	

Table 8: PFP-to-particle cross-attention ablation (1 M slices, 15 epochs, no Geant4 inputs), longest track per test slice.

3.4 The surrogated dataframes

`scripts/write_surrogate_df.py` turns the generated slices and PFPs into files in the production format, one `xsum_surrogate_N.df` per grid job N of the archive `prod_v10_14_02_03_official/`: `mcnu, mcprim, mcdaught, hdr` and the POT tables copied from the production; `slc_0` generated (index `(__ntuple,entry,rec.slc..index)`, `tmatch_idx` = generating neutrino, `tmatch_eff` / `tmatch_purity` NaN, the cohpi `pass_*` and `in_sr` flags recomputed from the generated variables, plus `surrogate_topology_class`); `pfp_0` generated (the 38 fields laid out under the production column names, fitted PFPs only, no induction-plane chi2 or truth-match columns); `provenance`; and `file_splits.parquet`, a copy of the generators' split (`--split-table`). Not in the files: cosmic-only and unmatched slices, PFPs without a fit, the induction-plane chi2 and the PFP truth-match columns.

The shipped dataset is `surrogate_official_m2vw_pfp` (2026-11-09), written so far for jobs 0–199 (6 GB) in `/pnfs/sbnd/persistent/users/lwan/dataframe/surrogate_official_m2vw_pfp/`; the other 1,800 jobs follow after review. Next to it, `surrogate_official_m3_pfp` (2026-11-05, 2,000 jobs, 60 GB) is the previous slice and PFP generators and remains the only complete dataset until then. Comparisons with the real chain must use the test files: the loader `dataframe/xsum_testsplit.py` reads `dataframe/file_splits.parquet` and the repaired `slc` sidecars and returns the real and the surrogated tables of the test files of any list of jobs. Analysers' comparisons so far: cohpi (S. Oh), kaon (T. Wester), CC1p0pi (M. Fernandez); the last is the acceptance test of Sec. 4.

4 Where we are: the CC1p0pi acceptance

The collaborator's CC1p0pi cut flow (`scripts/check_cc1p0pi_stages.py`, a transcription of their notebook) is run on the held-out test files of jobs 0–199 of the real production and of the surrogated dataset (180,000 neutrinos). The stages: a reconstructed vertex in the FV, `nu_score` > 0.45, exactly two prongs (two good PFP tracks), containment (the vertex and the start and end of both tracks inside the FV and on one side of the cathode), a track-score cut, a vertex-distance cut, a muon x cut and the muon/proton PID. Two signal definitions are used. The primary one (decided 2026-11-07): ν_μ CC with one muon, one proton and no pions, p_μ and p_p below 1 GeV, true vertex in the FV outside the cathode band, no end-point containment (7,659 signal neutrinos). The notebook's own definition requires in addition the true muon and proton end points inside one TPC's FV (2,382): that boundary is sharp and set by stochastic end points, so any surrogate with the end-point floor of Sec. 3.3 moves truth-contained signal out and truth-exiting events in, while the reconstructed

containment cut still selects the same mixture; it is kept as the second table because it is the analysers' definition.

Table 10 gives the result and Fig. 2 (second row) every stage with the signal and background differences. The shipped chain selects within 3% of Pandora at every stage up to the PID cut (5% there), reproduces the final efficiency (0.113 against 0.109) and purity (0.869 against 0.885), and is nearer Pandora than the previous slice generator at every stage: the background excess before the two-prong cut fell from +423 to +150 with the vertex residual. The remaining disagreement is at "2prong-contained": purity 0.403 against 0.425 under the primary definition (the previous chain 0.398; the test-sample precision is ±0.008) and 0.351 against 0.418 under the notebook definition. In events, the surrogate has 36 fewer primary signal and 135 more background neutrinos at that stage, and 203 fewer truth-contained signal neutrinos. Section 5 traces the two.

held-out files of jobs 0–199	selected	primary eff / pur	notebook eff / pur	final stage, primary
Pandora	3,555	0.197 / 0.425	0.623 / 0.418	0.109 / 0.885
shipped chain	3,654	0.192 / 0.403	0.538 / 0.351	0.113 / 0.869
previous slice generator, same PFP generator	3,720	0.193 / 0.398	0.537 / 0.344	0.108 / 0.875

Table 10: CC1p0pi at "2prong-contained" (selected, efficiency and purity under both signal definitions) and at the end of the selection (2prong-mup-PID).

5 What is wrong, and why

Figure 2 (third row) is the summary; this section gives the evidence. The diagnostics, all under `scripts/`, with their outputs and the full record of the studies in `reports/cc1p_containment/`:

- `diag_cc1p_containment.py`: per neutrino, the two longest good tracks of the Pandora and the surrogate slice, each containment condition separately, a generator-truth category;
- `diag_cc1p_topo.py`: PFP multiplicity against the exit of the longest track, per category;
- `diag_m2_link.py`, `diag_m2_vtx_dir.py`, `diag_m2_vtx_dirt.py`: the slice generator on its own test slices, the exit–multiplicity link and the reconstructed vertex by zone;
- `diag_contain_joint.py`: the containment conditions one by one and jointly.

Decomposition. At "2prong-contained" the primary signal is −36 and the background +135. A hybrid dataset, Pandora slice table with the generated PFPs (in-sample split), has background +31 and signal −31 and recovers the purity (0.426 against 0.435): the background excess comes through the slice block, the signal loss through the PFP generator. The background excess is in two categories: CC Nπ, whose two-prong slices pass containment 0.335 of the time against 0.290 (256 against 222 events), and NC, with 45 too many two-prong slices (1,392 against 1,347) at the right containment rate; CC other 0π and ν_e are within statistics.

5.1 The slice generator: the exit–multiplicity link of CC + pion slices

In Pandora, a CC Nπ slice whose muon leaves the detector has fewer PFPs: among slices with $n_{\mathrm{pfp}} = 3$ the longest track ends in the FV 0.418 of the time, against 0.56 at $n_{\mathrm{pfp}} \geq 5$ (`topo_m2vw.txt`). The generated slices show 0.498 at $n_{\mathrm{pfp}} = 3$, flat in n_{pfp}. Since 94% of $n_{\mathrm{pfp}} = 3$ slices are two-prong, the extra contained CC Nπ passes the cut. For 1μ1p the two-prong topology is 73% of the events,

so no tail is selected and the same defect does not show. At the slice level (`link_m2vw.txt`, truth CC + pion, reconstructed vertex in the FV, `nu_score`> 0.45) the link $P(n_{\mathrm{pfp}} = 3 \,|\, \mathrm{exit}) - P(n_{\mathrm{pfp}} = 3 \,|\, \mathrm{contained})$ is +0.025 in Pandora and +0.011 generated; without the vertex cut it is absent in both (+0.005), so it is carried by the reconstructed-vertex FV cut. Splitting by where the *true* vertex is (`vtx_dir_m2vw.txt`) separates two zones.

Zone 1, true vertex in the FV (30% of these slices), fixed. The reco vertex passes the FV cut for 0.983 (exiting muon) and 0.965 (contained) of the Pandora slices. With the vertex generated as an absolute position the generated core was 4–6 cm against 0.5 cm and exiting-muon events, whose true vertices lie nearer the downstream and side walls, were pushed out (pass 0.932). With the residual the displacement median is 0.4 / 0.6 cm (P(> 5 cm) 0.175 / 0.307 against Pandora 0.159 / 0.338, tails unchanged) and the pass rates are 0.970 / 0.974.

Zone 2, dirt neutrinos (61%), open. For interactions outside the active volume the reconstructed vertex sits on the true muon line inside the detector (at the entry wall for 0.31, off the line for 0.67) and the longest track runs from there to the far wall. The position of the reco vertex is reproduced; the track given the vertex is not (`vtx_dirt_m2vw.txt`): $P(\mathrm{exit} \,|\, \mathrm{reco\ vertex\ in\ FV})$ is 0.750 in Pandora and 0.627 generated (0.552 before the wall targets), and Pandora exiting tracks have a length equal to the path to the wall within 5% for 0.51 against 0.47. In the FV-pass table this zone reads 0.296 / 0.112 (Pandora, exit / contained) against 0.248 / 0.166, and it carries +0.10 of the +0.11 Pandora link.

Tried and ruled out (in-sample split, the same for every row; Fig. 8, Table 11). A two-stage flow with the PFP counts as dequantised numbers in stage 1 (M2s) closes the link in the chain but breaks the joint count lattice (17% of slices on tuples Pandora never populates) and loses two-prong efficiency; with the counts as a one-hot class in stage 1 (M2o) the counts are right and the link is lost; with the class from a classifier on the generated stage 1 (M2c) the counts and the acceptance equal the shipped generator and the link is still missing; sampling the topology after the flow (M2t), a 4× topology head and muon–hadron angles as extra inputs change nothing. None reproduces the link at the slice level (longest track in the FV for CC + pion $n_{\mathrm{pfp}} = 3$ slices 0.48–0.52 against 0.45). The topology head itself reproduces the Pandora link when fed the Pandora slice block and loses it on the generated block (`diag_m2_swap.py`): the link is carried jointly by the longest track, the presence of the optional groups and the counts, which the generator samples in separate pieces. The next lever is the dirt zone: the longest track generated jointly with the vertex (an exit predictor from the true muon line and its entry wall), or the wall relation imposed given the vertex.

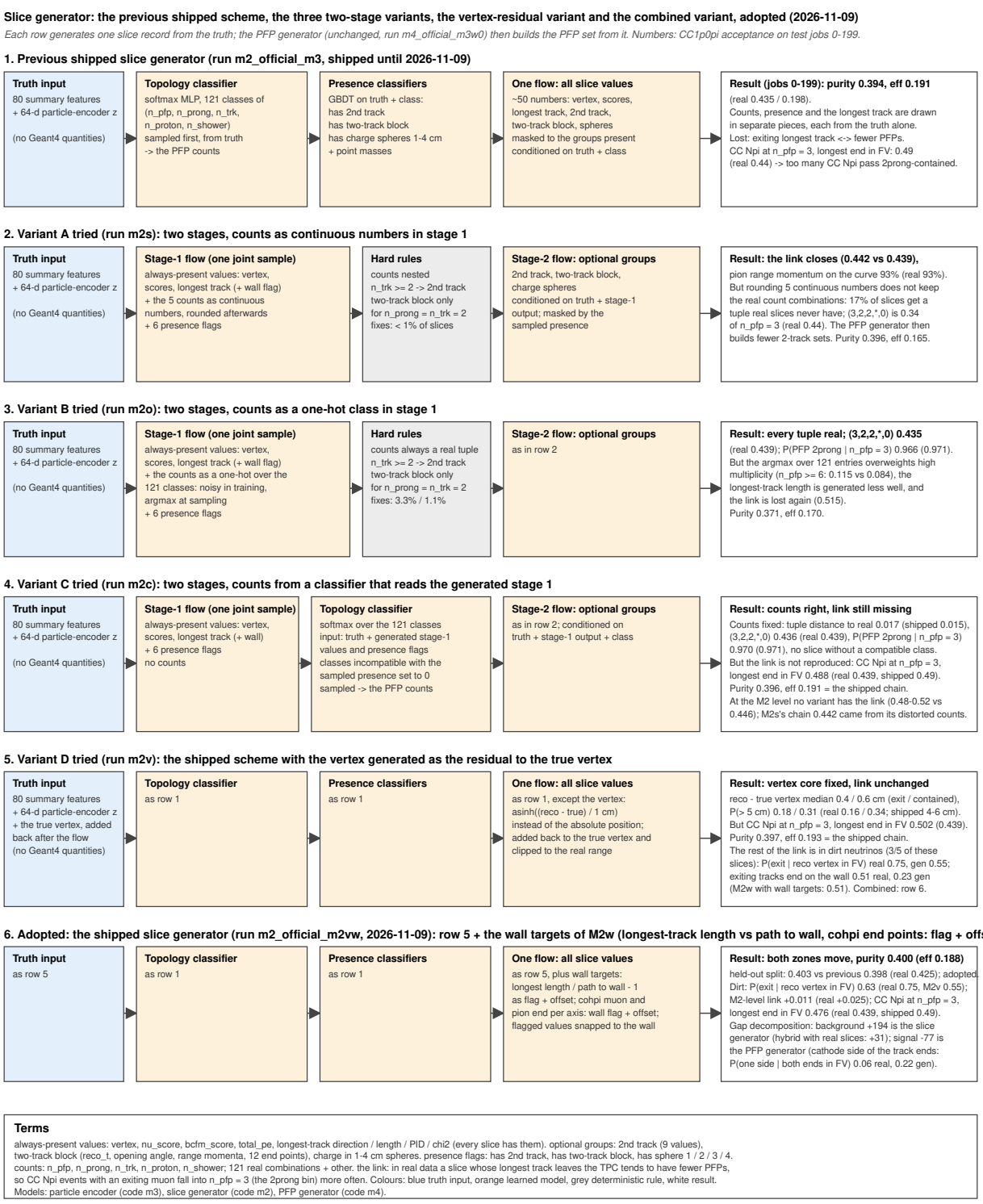


Figure 8: The slice generator: the previous shipped scheme, the three two-stage variants, the vertex-residual variant and the shipped combination, with what each changed (in-sample acceptance numbers).

run	change to the slice generator	n_3 end in FV	eff / pur
Pandora		0.439	0.198 / 0.435
previous (`m3`)	summary features + embedding	0.49	0.191 / 0.394
M2w	wall targets	0.491	0.185 / 0.390
M2t	topology sampled after the flow, range on curve	0.494	0.192 / 0.404
M2s	two stages, counts as numbers	0.442	0.165 / 0.396
M2o	two stages, counts as a one-hot class	0.515	0.170 / 0.371
M2c	two stages, class from a classifier	0.488	0.191 / 0.396
M2v	vertex as residual	0.502	0.193 / 0.397
shipped (`m2vw`)	vertex as residual + wall targets	0.476	0.188 / 0.400

Table 11: Slice-generator variants, all with the same PFP generator, on the in-sample split: the fraction of CC Nπ $n_{\text{pfp}} = 3$ slices whose longest track ends in the FV (n_3 end in FV) and the 2prong-contained efficiency / purity (test precision ± 0.008); the held-out numbers of the shipped generator are in Table 10.

5.2 The PFP generator: the cathode side of the track ends

Truth-contained signal neutrinos that both chains call two-prong pass containment 0.866 of the time in the surrogate against 0.984 in Pandora (-128 of 1,039; `diag_m2vw_summary.txt`); the condition that fails is the longest-track end in the FV (0.087 against 0.007) and, less, the cathode side (0.040 against 0.005). The mirror image is the primary signal whose true muon or proton leaves the FV: reconstructed as contained 293 times against 103. Each containment condition passes at the Pandora rate for these events (longest end in FV 0.236 against 0.224, second end 0.889 against 0.888, one cathode side 0.674 against 0.665), but their joint does not: given both track ends inside the FV, the vertex and the ends lie on one side of the cathode 0.064 of the time in Pandora and 0.220 generated (`contain_joint_m2vw_noclip.txt`). A Pandora muon that stops inside the FV after a long path has crossed the cathode; the generated end keeps the side of the vertex. The hybrid shows the same joint loss (0.205 against 0.086), so the defect is in the PFP generator, not upstream. It is a lost correlation, not resolution: the per-event end-point distance is at the two-draw floor, and the end-point models tried (a fix of the end-point bend, the end as a direct target with particle attention, two and four attention layers) left the FV and wall rates unchanged; Geant4 end-point inputs (M3g, M3w) restore the correlation but are not admissible. The lever is to generate the end relative to the cathode, a per-track flag "end on the other side of the cathode" with the crossing point, as a target of the set flow like the wall flags, or the end x as the signed distance to the cathode. This sets the notebook purity (0.351 against 0.418) and the -36 primary signal at containment.

5.3 Smaller items

- The cardinality head is split on the card table and saw 90% of the held-out files (Sec. 2).
- Range momenta: `range_p_mu` and `range_p_cpi` are functions of the track lengths (97% / 93% of Pandora slices within 2%), yet the flow generates them as independent numbers and smears the sharp range peak of stopping pions (peak fraction 0.17 against 0.22). Flagging them on the curve (M2t) reached 72% on the curve for the pion, the two-stage M2s 93% (Pandora 93%); not shipped.
- Not in the files: cosmic-only and unmatched slices, PFPs without a fit, the induction-plane chi2, the PFP truth-match columns.

- No uncertainty estimate per generated record yet, no physics holdouts yet.

6 Next steps

1. PFP generator: the cathode-side target of Sec. 5.2; check on the hybrid first.
2. Slice generator: the dirt-zone wall relation of Sec. 5.1.
3. Split every table by a hash of the file id and retrain the chain on it (Sec. 2).
4. Write the shipped dataset for all 2,000 jobs once reviewed (pass 2 of `m2vw_pfp`); then retire the previous complete dataset (`m3_pfp`).
5. Range momenta from the generated lengths, with the slice generator's version of the two-track figure.
6. PFP completeness: induction-plane chi2, PFPs without a fit, truth-match columns; cosmic-only and unmatched slices from the MC.
7. Uncertainty: Bayesian last layers or ensembles, an out-of-support flag per generated record.
8. Holdouts: train on $|t| < 0.1$, test the tail; low versus high E_ν; forward versus wide-angle muons; ≤ 2 versus ≥ 3 charged primaries; one TPC versus the other.
9. Collaborators' comparisons (cohpi, kaon, CC1p0pi) on the test files of `surrogate_official_m2vw_pfp`.

A The path to the shipped version

Kept for traceability; the numbers are those recorded at the time (in-sample split before 2026-11-09).

Baselines (M1). Gradient-boosted classifiers and mixture density networks on the 80 summary features set the floor for every target: slice efficiency AUC 0.977 (32% of all true neutrinos have a matched slice), PFP multiplicity accuracy 56% against 32% marginal, and conditional marginals of the vertex, leading-track and two-track variables. The code remains (`xsum.train.baselines`).

Slice generator v1–v3. v1 (unbounded cosines, counts as dequantised flow dimensions) closed the cohpi cut flow only to 45% in the signal region; v2 (bounded cosines, point-mass heads) to 74%, losing most at the exact-count stage; v3 (topology head, 2026-10-22) to better than 10% at every stage. v3 with the particle-set embedding (`m2_official_m3`, 2026-11-05) fixed the per-event directions (Table 2; CC1p0pi two-prong-contained efficiency 0.387→0.543 under the notebook definition); the vertex residual and the wall targets (`m2_official_m2vw`, 2026-11-09) completed it (Sec. 5.1).

PFP generator v2 to M3w0. v2 (29 fields, end as start + length × direction) shipped with `surrogate_official_v3_pfp`; the particle-set inputs (M3) and then the generated end point with the per-axis wall flag and offset (layout v4, M3w0, 2026-11-08) followed. Models tried on the end point: M3b (a fix of the end-point bend) and M3e (the end as a direct target plus cross-attention) left it unchanged; M3g (Geant4 start, end, length and containment on the

particle tokens; error 20→7 cm) and M3w (Geant4 straight-line exit point and wall flag) are not admissible and were kept as diagnostics only. The in-sample acceptance at two-prong-contained: M3 0.199 / 0.397, M3w 0.190 / 0.395, M3w0 0.191 / 0.394, hybrid (Pandora slices + M3w0 PFPs) 0.194 / 0.426, Pandora 0.198 / 0.435.

Datasets. `surrogate_v3_6ch` (scratch production, slices only), `surrogate_official_v3` (2026-10-22, slices only, 2,000 jobs), `surrogate_official_v3_pfp` (first dataset with every reconstructed object surrogated) and `surrogate_official_m3w0_pfp` (jobs 0–199) are deleted (2026-11-09); `surrogate_official_m3_pfp` remains until the shipped dataset is complete.

Signal definition. The notebook's truth-contained signal was replaced as the primary definition on 2026-11-07 (Sec. 4); the test-split finding and fix are dated 2026-11-09 (Sec. 2).